Novelty audit: the midpoint contrast and the anchor × template interaction

Subject. *Changing One Word* (v6), frozen at `releases/v6-final/`. **Status of the work audited.** An exploratory controlled extension of the forward–backward asymmetry literature, not a discovery claim. **Question.** Which parts of that study are already established, which are specific to our design, and what remains unresolved. **Sources.** Primary PDFs only, page-anchored in `EVIDENCE.md`: Zhang et al. (arXiv:2605.07127), Sclar et al. (ICLR 2024, arXiv:2310.11324), Leidinger et al. (Findings EMNLP 2023), plus Liu et al. (*Lost in the Middle*) and Berglund et al. (*Reversal Curse*) as adjacent results. **Date of search.** 2026-10-27.

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1. Summary

1. **The direction asymmetry is not ours.** Zhang et al. establish that backward positional retrieval lags forward retrieval across model families, anchors, query types and item types (p1–2, p5). Our baseline cells replicate that asymmetry.
2. **Prompt sensitivity is not ours either.** Sclar et al. show formatting changes worth up to 76 accuracy points on 53 classification and multiple-choice tasks (p1, p5); Leidinger et al. show grammatical and lexical variation worth up to 5 points on QA and NLI (p1, p5). Neither concerns ordinal indexing.
3. **The midpoint queries are already in Zhang's protocol, at L=5.** Their operators are `End+ : S[n]` and `End− : S[−n]` (p3) instantiated over the valid offsets (p13), so at L=5 the ordinal n=3 makes the two queries address the same item. What the paper does not report is an analysis built on that coincidence: the results are condition-level aggregates, the published methods do not document pairing, and no template manipulation is reported. Our contribution is the explicit same-list, same-item, same-ordinal *analysis*, plus the anchor × template estimate — not the existence of the queries.
4. **The interaction is motivated by prior prompt-sensitivity research.** That an anchor effect might depend on the question template follows from Sclar and Leidinger at the level of "templates matter"; we did not find a within-item estimate in positional retrieval in the sources reviewed. Ours uses one paraphrase, so it quantifies an interaction between these two templates, not a general law.
5. **What remains genuinely open** is narrow: mechanism, behaviour at larger scales, and whether the midpoint difference is anchor-specific or an instance of template noise. Our data settle none of these.

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2. Comparison matrix

2.1 Established findings we inherit

#	Finding	Source, page	Scope of the evidence
E1	Backward positional retrieval substantially lags forward retrieval	Zhang p1–2, p5; Fig. 2c	2 query types × 2 directions (four tasks) × 2 anchors × 2 item types, L ∈ {5,10,20}; open and closed models
E2	The gap is not explained by counting	Zhang p3, §3.1; Fig. 2 row 1	Counting near-ceiling (78–100%) on the same short lists
E3	Per-position accuracy curves are available for both directions	Zhang p18, Fig. 10; p14, Fig. 7	L=20, backward panels ordered by absolute position
E4	Position of evidence in a long context gives a U-shaped curve	Liu p1	Content-based retrieval, multi-document QA and KV retrieval
E5	Prompt *formatting* changes accuracy by tens of points	Sclar p1, p5, p8	53 Super-NaturalInstructions tasks: 19 multiple-choice, 34 classification
E6	Semantically equivalent *grammatical/lexical* variation changes accuracy	Leidinger p1, p5	BoolQ, RTE, CB, ARC-E, SST, IMDB; drops up to 5 pp
E7	Direction asymmetry also appears in parametric factual recall	Berglund p1	"A is B" vs "B is A" after training, not in-context indexing

2.2 Our exact experimental differences

#	Our design choice	What it changes relative to the sources	Where it matters
D1	**Same list, same item, same ordinal.** At odd L the midpoint satisfies p = L−p+1, so "k-th from the beginning" and "k-th from the end" are the same query up to the anchor word	Zhang's protocol contains both midpoint queries only at L=5 (n=3); elsewhere the two directions address different items at the same ordinal, and different ordinals for the same item	Makes the anchor the only difference between the two arms of a pair (E1)
D2	**Within-list pairing.** Each item contributes a forward and a backward outcome	The published methods do not document cross-condition pairing, and the paper does not report a paired midpoint analysis. (p13)	Holds list identity constant and enables a paired statistic (Newcombe, exact McNemar); the precision gain depends on within-pair covariance; see §2.4
D3	**Odd lengths only** (7, 9, 11)	Zhang's grid is 5, 10, 20 — the midpoint ordinal exists only at L=5	The contrast is expressible at every length we run
D4	**Two templates × two anchors, crossed**	Zhang randomises phrasing only in SFT data (p13); evaluation uses a fixed template per condition	Allows an interaction estimate rather than a main effect
D5	**Forced-choice scoring with generation cross-check**	Zhang uses 3-shot generation, temperature 0.7, exact match	Removes decoding noise and parser sensitivity from the contrast; both modes gave identical 2×2 tables
D6	**Nine cells, one position, 24 items, uncorrected**	Much smaller than any source here	Bounds what can be claimed

2.3 Genuinely unresolved novelty

#	Claim	Status after the audit	Why it is not already covered
N1	A **paired, same-item, same-ordinal** estimate of the anchor effect	Not reported in any source we could find	Requires D1 + D2. Zhang's protocol contains the two midpoint queries at L=5, but the paper does not report a paired midpoint analysis
N2	The estimated anchor effect **differs between the two tested templates**, measured within item (−54.17 pp, exact p = 2/4096)	Not reported in positional retrieval; measured here	Sclar and Leidinger establish template sensitivity in other task families, but not an anchor × template interaction in positional retrieval
N3	Mechanism of the midpoint difference	Open	Zhang explicitly speculates (p8, "We speculate…"); no test in either paper
N4	Behaviour at larger models and longer odd lengths	Open	Our cells are 0.5B and 1.5B at L ≤ 11; Zhang's models are larger but his grid has no odd length above 5

2.4 On independent samples and pairing

Independent and paired designs estimate the same population contrast only under matched experimental conditions: the same position set, the same aggregation rule, the same item pool. That condition does not hold between the two studies here. Zhang's reported quantity averages over all valid offsets, so its estimand is a position-averaged anchor effect; ours is the anchor effect at the midpoint only. These are different population quantities, and the difference is one of estimand rather than of statistical efficiency. A design that samples lists independently and one that pairs lists within items both target an all-position quantity if they use the same offsets and aggregation rule; neither equals a midpoint-only contrast.

Within that caveat, independent samples are a valid way to estimate a contrast. The published methods do not document cross-condition pairing, and the paper does not report a paired midpoint analysis; nothing in this audit verifies list reuse from code or data. Our design does not show that theirs is wrong, and their aggregate is not invalidated by the absence of a reported pairing.

Pairing holds list identity constant and enables paired analysis; its precision benefit depends on within-pair covariance. When the two arms of a pair are strongly correlated, the paired difference has smaller variance than the difference of independent means; when they are not, the benefit shrinks. What pairing guarantees is the design property — the anchor is the sole within-pair difference, with item-level tests (exact McNemar, Newcombe) available — not a given gain in precision.

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3. What the matrix rules out claiming

- **Not "first to show backward indexing is hard."** E1 is Zhang's contribution, replicated by us.
- **Not "first to show prompts matter."** E5 and E6 predate this work, with larger effects and larger task suites.
- **Not "the effect is a positional artefact."** Our own v6 retracted that; Zhang evaluates all valid offsets, so their aggregate is position-balanced even though the paper does not report a paired midpoint analysis.
- **Not "the interaction isolates digit-versus-word ordinals."** The paraphrase also changes "What" to "Which" and drops "at the … position". Sclar's grammar (B2) is a reminder that several formatting dimensions move together unless crossed.
- **Not "Zhang's design is inadequate."** Independent samples can estimate either a midpoint-only or an all-position contrast. The estimand depends on the positions and aggregation used; pairing determines the relationship between observations. The differences between the studies are estimand and pairing structure, not validity.
- **Not "first".** No search establishes priority, and this audit does not attempt to.

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4. Search limitations

Missing citations, absent search hits and failed queries are limitations of this audit, not evidence of originality. Semantic Scholar returned zero citations for the Position Curse and 100 (sampled) for Sclar; arXiv API searches for `abs:"position-based retrieval"` returned only the Position Curse itself. Google Scholar, Scopus and ACL Anthology full-text search were not reachable from this host, and Semantic Scholar lags recent arXiv deposits. A zero citation count reflects indexing lag as much as the state of the literature, and this audit draws no inference from it.

The statements in §2.3 are accordingly of the form "not reported in any source we could find" and "the paper does not report" — not "does not exist" and not "was never done". Nothing here is a priority claim.

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5. What a reviewer could still object to, and the answer

Objection	Answer	Where it is answered
"Fig. 10 already compares forward and backward at matched positions."	Those panels order by *absolute position*; at position p the forward query names ordinal p and the backward query names L−p+1. The published methods do not document cross-condition pairing, and the paper does not report a paired midpoint analysis, so it is a visual comparison rather than an item-level contrast.	EVIDENCE A3, A4
"Independent samples are equally valid."	Yes. Independent samples can estimate a midpoint-only or an all-position contrast. Zhang's reported all-position aggregate differs from our midpoint-only estimand; pairing separately holds list identity constant and enables paired analysis.	§2.4
"Prompt sensitivity is well known."	Yes — for formatting (Sclar) and grammar (Leidinger), in classification tasks. It is not reported in the sources reviewed whether the estimated anchor effect differs between templates in positional retrieval; across the two templates we tested, it does.	EVIDENCE B1–B3, C1–C4
"One paraphrase is not a template sample."	Correct, and stated as a limit. It supports an interaction estimate between these two templates, not a distribution over templates.	v6 §5
"24 items per cell."	Correct. The intervals are wide by construction, and the margin verdicts reflect that.	v6 §3.1, §5
"The effect could be a length effect, since pools differ."	Length is confounded with pool (J–P / J–R / J–T), stated in v6 §2.3; only the within-list contrasts are used for claims.	v6 §2.3, §5

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6. Contribution statement

An exploratory controlled extension of the forward–backward asymmetry literature: a same-list, same-item, same-ordinal paired estimate of the anchor effect in one small model, and evidence that the estimated effect differs between the two question templates tested. The asymmetry itself, and prompt sensitivity in general, are prior results. This study adds the paired midpoint analysis and its within-item interaction estimate, at the scale and with the limits reported in v6, and makes no priority claim.